

Vagally Mediated Heart Rate Recovery After Exercise Is Accelerated in Athletes but Blunted in Patients With Chronic Heart Failure

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Objectives. Vagally mediated heart rate recovery after exercise was assessed in patients with chronic heart failure and in well trained athletes by analyzing the postexercise heart rate decay.

Background. Vagal reactivation is an important cardiac deceleration mechanism after exercise. However, alterations of this mechanism under pathologic conditions have not been characterized because of the lack of a specific index.

Methods. To find a vagally mediated component of heart rate recovery, the time constants of the beat-by-beat heart rate decay for the first 30 s (T_{30}) and the first 120 s (T_{120}) after exercise were obtained at six levels of exercise in eight normal volunteers: 1) at maximal exercise, 2) at anaerobic threshold, 3) at anaerobic threshold with propranolol administration, 4) at anaerobic threshold with atropine administration, 5) at anaerobic threshold with concomitant administration of both drugs, and 6) at 50% of anaerobic threshold. To investigate the effects of heart failure and endurance training on vagally mediated heart rate recovery, T_{30}

and T_{120} at anaerobic threshold were obtained in 20 patients with chronic heart failure and in 9 cross-country skiers.

Results. In normal volunteers, T_{30} and T_{120} were markedly prolonged by atropine administration, indicating that both time constants are mediated by vagal reactivation. Moreover, T_{30} was almost independent of the exercise intensity and sympathetic blockade, whereas T_{120} was affected by sympathetic nerve activity and exercise work load. These results indicate that T_{30} is mediated primarily by vagal reactivation, independent of sympathetic withdrawal, and is significantly smaller in athletes ($p < 0.01$) and significantly larger in patients with chronic heart failure ($p < 0.01$) than that in respective age-matched normal control subjects.

Conclusions. The T_{30} value could be a specific index for vagally mediated heart rate recovery. Vagally mediated heart rate recovery after exercise is accelerated in well trained athletes but blunted in patients with chronic heart failure.

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During exercise, heart rate increases through a combined effect of sympathetic activation and vagal withdrawal. These autonomic changes are mediated by both somatic exercise reflexes and central command mechanisms (1-3). By contrast, the increased heart rate rapidly declines after cessation of exercise (4,5). This rapid recovery appears to be an important mechanism for avoiding excessive cardiac work after exercise. Although autonomic nerve mechanisms of the rapid heart rate recovery have not been fully understood, several lines of evidence have suggested that vagal reactivation plays an important role (6-8). However, despite the physiologic importance, alterations of vagally mediated heart rate recovery under pathologic conditions have not been extensively studied, most likely because of the lack of a specific index. Power spectral analysis of heart rate is a useful technique for detect-

ing changes in vagal activity (9,10). However, this technique is inappropriate for evaluation of dynamic and rapid vagal reactivation because of limited time resolution. Analysis of the beat-by-beat heart rate decay after exercise may have the potential to deduce an indicator of vagally mediated heart rate recovery, providing fine time resolution.

Although autonomic nerve mechanisms are unclear, heart rate recovery after exercise is delayed in patients with chronic heart failure (8). These patients are reported to have altered vagal activity (11-14). Thus, the delayed heart rate recovery in patients with chronic heart failure may be attributed to impaired vagal reactivation after exercise. In contrast, endurance physical training is associated with accelerated heart rate recovery after exercise (15,16). Although autonomic mechanisms of this acceleration are also unclear, well trained athletes are known to have an increased tonic vagal activity (17). Thus, postexercise vagal reactivation may be augmented in athletes, allowing for rapid heart rate recovery.

The purpose of the present study was therefore twofold: 1) to derive a vagally mediated component from the beat-by-beat heart rate decay after exercise by using pharmacologic autonomic nerve blockade in normal volunteers; and 2) to investigate alterations of vagally mediated heart rate recovery

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Table 1. Clinical Characteristics of Study Groups

	Age-Matched Control Group for CHF (n = 20)	CHF Group (n = 20)	Age-Matched Control Group for Athletes (n = 11)	Athlete Group (n = 9)
Age (yr)	56 ± 6	55 ± 12	20 ± 4	20 ± 2
LVEDD (mm)	50 ± 4	62 ± 10*	46 ± 5	52 ± 2†
LVESD (mm)	28 ± 5	52 ± 11‡	30 ± 3	35 ± 3†
FS (%)	43 ± 8	17 ± 6‡	35 ± 2	32 ± 3

*p < 0.05, †p < 0.01 versus control group for chronic heart failure (CHF).
‡p < 0.05 versus control group for athletes. Data are expressed as mean value ± SD. FS = fractional shortening; LVEDD (LVESD) = left ventricular end-diastolic (end-systolic) dimension.

in patients with chronic heart failure and in well trained athletes by using the derived index.

Methods

Subjects. To extract a vagally mediated component from the postexercise heart rate decay, eight sedentary normal male volunteers, ranging in age from 28 to 39 years (mean 34), were subjected to pharmacologic autonomic blockade challenges. All subjects had a normal physical examination, including a rest and exercise ECG. None had a history of cardiovascular diseases, and none was taking medications during the study.

To investigate changes in vagally mediated heart rate recovery in patients with chronic heart failure and in athletes, we studied 20 men with chronic heart failure (the heart failure group) and 9 male cross-country skiers trained for at least 2 years (the athlete group). Twenty and 11 normal sedentary men formed two age-matched control groups for the heart failure and athlete groups, respectively. Clinical characteristics of each group are shown in Table 1.

The heart failure group included 14 patients with dilated cardiomyopathy and 6 with a previous myocardial infarction. All patients were in New York Heart Association functional class II. Left ventricular fractional shortening determined with echocardiography ranged from 7% to 25% (mean 17%). All patients in this cohort had been treated with cardiovascular drugs for >3 months before the study (diuretic agents in 13, digitalis in 5, nitrates in 7, nifedipine in 9, angiotensin-converting enzyme inhibitors in 4, mexiletine in 7). These medications were continued throughout the study without alterations in the dose or regimen. No patients had been treated with beta-adrenoceptor acting agents, diltiazem or disopyramide.

In the athlete group, end-diastolic left ventricular dimension was slightly but significantly larger than that in the control group for the athlete group. No subjects had any history of cardiovascular diseases, and all had a normal physical examination and ECG.

The age-matched control subjects for the heart failure and athlete groups were selected from patients who had been screened for cardiac diseases at our hospital. Organic cardio-

vascular abnormalities were excluded by routine examinations, including ECG, echocardiographic and exercise testing.

All subjects were in sinus rhythm. Written informed consent was obtained from all subjects before the study.

Determination of postexercise heart rate decay. A limb-lead ECG was recorded continuously from rest to 120 s after cessation of exercise. The timing signal of cessation of exercise was simultaneously fed into a personal computer (NEC Co. model PC-9801, Tokyo, Japan) equipped with an analog/digital converter. The analog data of the RR intervals, excluding ectopic beats, were transferred to a series of RR intervals measured with a precision of 0.001 s. The beat-by-beat heart rate and the elapsed time from cessation of exercise were then calculated from the series of RR intervals. Thus, the beat-by-beat heart rate decay for 120 s after exercise was automatically obtained.

Protocol in normal volunteers. To examine the dependence of the postexercise heart rate decay on autonomic nerve activities and exercise intensity, the beat-by-beat heart rate decay was obtained at the following six exercise levels in eight normal volunteers: 1) at maximal exercise, 2) at anaerobic threshold, 3) at anaerobic threshold with intravenous administration of 0.2 mg/kg body weight of propranolol, 4) at anaerobic threshold with intravenous administration of 0.02 mg/kg of atropine sulfate, 5) at anaerobic threshold with concomitant administration of both drugs, and 6) at 50% of anaerobic threshold. The exercise tests were performed in this order at 1-week intervals.

Maximal upright bicycle exercise was performed first. After 5 min of rest on the upright ergometer, exercise was conducted with 1-min unloading pedaling followed by 20-W incremental loading every minute until subjects were nearly exhausted. Gas exchange data were collected throughout exercise using a breath-by-breath respirometry system (Minato Co. model RM-200, Tokyo, Japan). The system was calibrated immediately before each study using a 2-liter calibration syringe and a gas mixture of 14.93% oxygen, 5.00% carbon dioxide and 80.07% nitrogen. Oxygen uptake, carbon dioxide output, minute ventilation and respiratory gas exchange ratio were obtained every 10 s during exercise. Anaerobic threshold was defined as the time point when the ratio of minute ventilation to oxygen uptake began to increase rapidly without a concomitant increase in the ratio of minute ventilation to carbon dioxide output (18). The work load at the point of anaerobic threshold was also determined to standardize the exercise intensity during subsequent submaximal exercise tests. Systolic blood pressure was measured every minute using a sphygmomanometer throughout exercise. The heart rate decay for 120 s after cessation of exercise was then obtained.

A 4-min single-stage exercise test was used for the remaining five submaximal exercise tests. During each exercise, the work load was kept constant for 4 min, and the postexercise heart rate decay was then obtained. Gas exchange data and systolic blood pressure were also measured every 10 and 60 s, respectively. Medications for autonomic blockade were administered over 3 min after a 5-min rest period on the ergometer. Exercise was begun 3 min after the injection had been completed.

Protocol in the heart failure and athlete groups. The beat-by-beat heart rate decay for 120 s after exercise at anaerobic threshold was obtained in the heart failure group, the athlete group and their respective control groups. To determine anaerobic threshold as well as peak oxygen uptake in individual subjects, maximal bicycle exercise was preliminarily performed with the same time intervals and measurements as those used in the normal volunteers. The incremental loading was set at 10 W/min in the heart failure group and 20 W/min in the remaining three groups. One week after the maximal exercise test, the 4-min single-stage exercise test with the work load corresponding to anaerobic threshold was performed to obtain the postexercise heart rate decay.

Reproducibility of heart rate decay. Reproducibility of the postexercise heart rate decay was tested in 5 of 8 normal volunteers subjected to autonomic nerve blockade and in 11 heart failure group patients. In normal volunteers, exercise at anaerobic threshold without autonomic blockade was performed again 1 week after the last exercise test. The interval between the two exercise tests at anaerobic threshold was therefore 5 weeks. In the patients with chronic heart failure, the exercise study was performed again within 3 months after the first study (mean interval 44 days). During this period, exacerbation or improvement of heart failure was not observed in any patient.

In this study all exercise tests were performed during a postabsorptive state between 1 and 5 pm in the air-conditioned room at 25°C.

Analysis of heart rate decay. It has been demonstrated that heart rate decreases monoexponentially after exercise (6,19). Perini et al. (6) suggested that the time constant of the heart rate decay for 5 min after exercise depends on both rapid vagal reactivation and gradual sympathetic withdrawal. Therefore, the time constant of an earlier phase of the heart rate decay may predominantly reflect the rapid vagal reactivation. In this study the time constants of the heart rate decay for the first 120 s (T_{120}) and the first 30 s (T_{30}) after exercise were computed as candidates for the index of vagally mediated heart rate recovery. The T_{120} time constant was obtained from dynamic analysis of the heart rate decay by adopting the following equation using a computer program (Sigma Plot 4.10, Jandel Scientific):

$$\text{Heart rate (t)} = a_0 + b \cdot \exp(-t/T_{120}),$$

where a_0 = asymptotic value of heart rate, and b = the decrement below the heart rate value at the end of the exercise for $t = \infty$. However, this technique cannot be used to compute T_{30} because of possible errors in fitting a curve to the steep section only of the heart rate decrease during the initial 30 s and determining a_0 from this section. Thus, a simple semilogarithmic regression technique was used to determine T_{30} (see Results).

Statistical analysis. Results are expressed as mean value $\pm$ SD. Intragroup comparisons were performed using analysis of variance with repeated measurements. Intergroup comparisons were performed using analysis of variance and the Scheffé

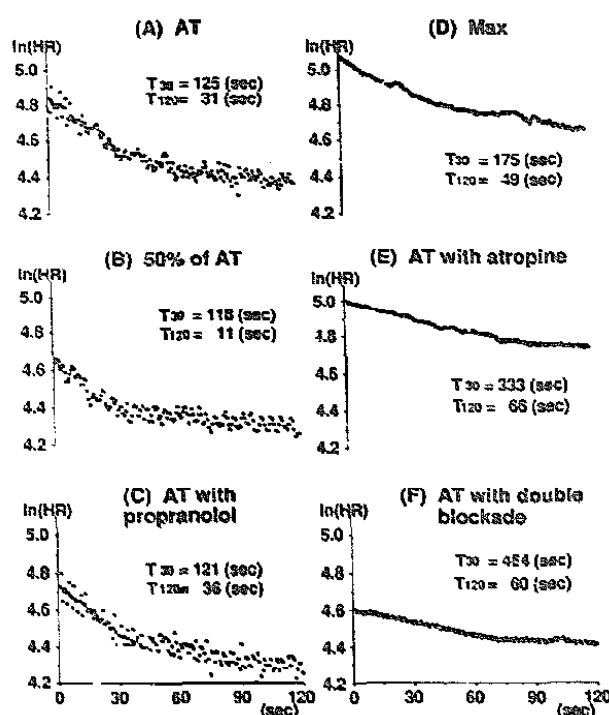


Figure 1. Postexercise heart rate decay at six exercise levels in a representative normal volunteer. A, At anaerobic threshold (AT). B, At 50% of anaerobic threshold. C, At anaerobic threshold with propranolol. D, Maximal exercise (Max). E, At anaerobic threshold with atropine. F, At anaerobic threshold with atropine plus propranolol. HR = heart rate; T_{30} , T_{120} = time constants of the heart rate decay for 30 s and 120 s, respectively.

test or nonpaired t test, as appropriate. Linear regression analysis was performed by the least-squares method; $p < 0.05$ was considered significant.

Results

Autonomic blockade challenges in normal volunteers. Respiratory gas analysis. In eight normal volunteers, the peak oxygen uptake and oxygen uptake at anaerobic threshold ranged from 33.6 to 44.5 ml/min per kg (mean 36.3) and from 26.6 to 31.0 ml/min per kg (mean 29.3), respectively. The work load corresponding to anaerobic threshold ranged from 152 to 192 W (mean 173).

Semilogarithmic analysis of heart rate decay. In the normal volunteers, when the natural logarithm of heart rate was plotted against the elapsed time after exercise, two linear components (an initial rapid decrease and a subsequent slow decrease) were observed at four exercise levels without atropine administration (Fig. 1, A to D). The break points of 32 decay curves obtained from the four levels in eight subjects occurred between 20 and 40 s after exercise. In contrast, the initial rapid decrease disappeared with atropine or atropine plus propranolol administration (Fig. 1, E and F), suggesting

Table 2. Hemodynamic Data and Oxygen Uptake at Rest and at the End of Exercise and Time Constants of Postexercise Heart Rate Decay at Six Exercise Levels in Normal Volunteers

	50% of AT	AT	AT With Propranolol	AT With Atropine	AT With Atropine Plus Propranolol	Max
Rest HR (beats/min)	77 ± 16	78 ± 18	61 ± 11*	92 ± 18	77 ± 8	76 ± 16
Rest SBP (mm Hg)	120 ± 7	121 ± 6	120 ± 6	118 ± 6	118 ± 7	121 ± 8
Exercise HR (beat/min)	102 ± 11*	139 ± 13	114 ± 9*	156 ± 10*	120 ± 7*	169 ± 9*
Exercise SBP (mm Hg)	125 ± 8*	176 ± 18	145 ± 6*	175 ± 11	158 ± 9*	191 ± 15*
Exercise VO ₂ (ml/min per kg)	15.3 ± 1.1*	29.3 ± 1.8	28.5 ± 2.1	28.6 ± 2.0	29.0 ± 2.7	36.3 ± 2.4*
T ₃₀ (s)	159 ± 56	138 ± 30	147 ± 30	357 ± 126*	348 ± 81*	174 ± 33*
T ₁₂₀ (s)	18 ± 9*	33 ± 13	34 ± 14	74 ± 17*	53 ± 13*†	60 ± 20*

*p < 0.05 versus anaerobic threshold (AT). †p < 0.05 versus anaerobic threshold with atropine. Data are expressed as mean value ± SD. HR = heart rate; Max = maximal exercise; SBP = systolic blood pressure; T₃₀, T₁₂₀ = time constants of postexercise heart rate decay for 30 and 120 s; VO₂ = oxygen uptake.

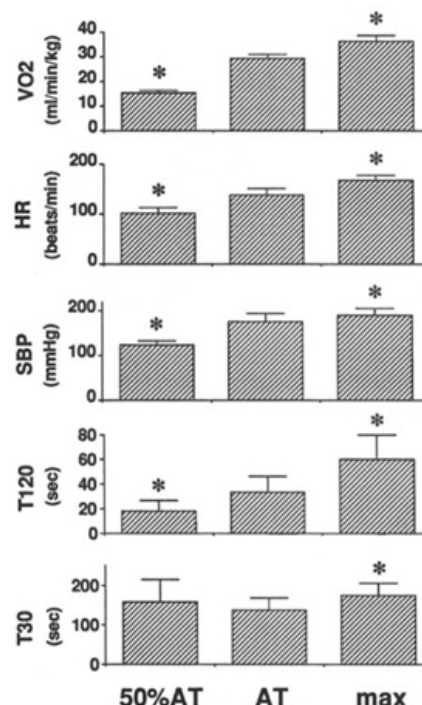
that this component depends on vagal reactivation. To find the best fit for the initial rapid decrease, linear regression analysis was performed on three sections of the 32 individual heart rate decay curves; sections analyzed were from 0 to 20, 0 to 30, and 0 to 40 s in the elapsed time after cessation of exercise. The slope of the regression line was almost identical between the sections from 0 to 20 and from 0 to 30 s but significantly less steep in the section from 0 to 40 s (0 to 20 s, -7.8 ± 1.8 ; 0 to 30 s, -7.6 ± 1.6 ; 0 to 40 s, $-6.0 \pm 1.4 \times 10^{-3}/s$; 0 to 30 s vs. 0 to 40 s, $p < 0.05$). Moreover, the correlation coefficient of the regression line for the section from 0 to 30 s was highest (0 to 20 s, 0.85 ± 0.12 ; 0 to 30 s, 0.89 ± 0.05 ; 0 to 40 s, 0.88 ± 0.06). Therefore, as an index of the initial rapid decrease, the time constant of the postexercise heart rate decay for the first 30 s (T₃₀) was determined as the negative reciprocal of the slope of the regression line.

Work load dependence of heart rate decay. The work load dependence of T₃₀ and T₁₂₀ was assessed using three discrete levels of exercise (i.e., at 50% of anaerobic threshold, at anaerobic threshold and at maximal exercise). Oxygen uptake, systolic blood pressure and heart rate at the end of exercise were increased with increasing work load. A similar trend was noted in T₁₂₀, indicating that this index largely depends on the exercise work load. In contrast, T₃₀ was nearly independent of exercise intensity; although the value at maximal exercise was significantly greater than that at anaerobic threshold, the difference was slight (Table 2, Fig. 2).

Autonomic nerve dependence of heart rate decay. The autonomic nerve dependence of T₃₀ and T₁₂₀ was assessed from their responses to autonomic blockade during exercise at anaerobic threshold. Oxygen uptake at the end of exercise was not affected by autonomic blockade. Compared with the value during the baseline exercise, exercise heart rate was significantly increased by atropine administration but was decreased by propranolol or concomitant administration of both drugs. Similar trends were noted in systolic blood pressure. Both T₃₀ and T₁₂₀ were unaffected by propranolol, but they were markedly prolonged by atropine administration (Table 2, Fig. 3). Moreover, there was a significant correlation between T₃₀

and T₁₂₀ ($n = 24$, $r = 0.64$, $p < 0.0001$). These findings indicate that both T₃₀ and T₁₂₀ are vagally mediated. However, prolongation of T₁₂₀ with atropine was significantly attenuated by concomitant administration of propranolol (Fig. 3), suggesting that T₁₂₀ also depends on sympathetic activity. This effect of propranolol was not observed at T₃₀. Thus, in contrast to T₁₂₀, T₃₀ was mediated primarily by vagal activity, independent of sympathetic activity.

Alterations in heart rate decay in the heart failure and athlete groups. Peak oxygen uptake in the heart failure group (range 14.2 to 21.7 ml/min per kg) was significantly less than

Figure 2. Work load dependence of postexercise heart rate decay. Data are expressed as mean value ± SD. SBP = systolic blood pressure; VO₂ = oxygen uptake; other abbreviations as in Figure 1. *p < 0.05 versus anaerobic threshold.

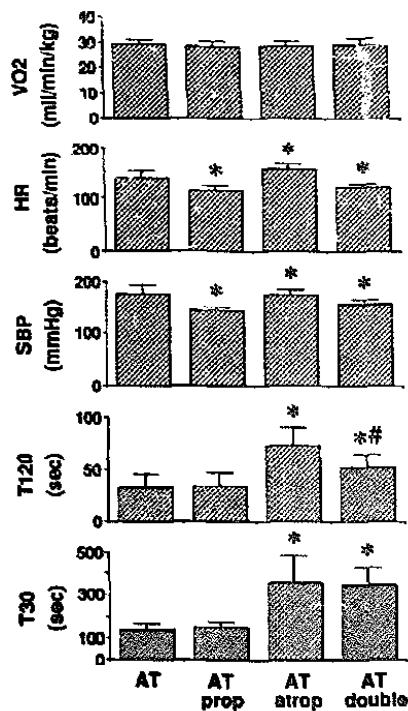


Figure 3. Autonomic nerve dependence of postexercise heart rate decay. Data are expressed as mean value $\pm$ SD. atrop = atropine sulfate; double = atropine plus propranolol; prop = propranolol; other abbreviations as in Figures 1 and 2. * $p < 0.05$ versus anaerobic threshold. # $p < 0.05$ versus anaerobic threshold with atropine.

that in the age-matched normal control group, whereas that in the athlete group (range 60.0 to 71.6 ml/min per kg) was significantly greater than that in the control group for athletes (Table 3). A similar trend was observed in oxygen uptake at anaerobic threshold.

To assess vagally mediated heart rate recovery, T_{30} and T_{120} were determined after exercise at anaerobic threshold in individual subjects from the four study groups. Both T_{30} and T_{120} were

significantly less in the athlete group and significantly greater in the heart failure group than in the respective age-matched normal control groups (Table 3). These changes in the time constants were independent of heart rate at the end of exercise, although rest heart rate was significantly slower in the athlete group than in the control group for athletes (Table 3).

Reproducibility of time constants. The variability of T_{30} and T_{120} was tested by repeated exercise tests at anaerobic threshold in 5 normal volunteers and 11 patients with heart failure. The standard deviation for T_{30} was 25 s ($n = 16$, $r = 0.92$, $y = 9.8 + 0.94x$, $p < 0.0001$), and that for T_{120} was 5 s ($n = 16$, $r = 0.95$, $y = 1.1 + 0.99x$, $p < 0.0001$). These results indicate that the time constants of postexercise heart rate decay are fairly reproducible over several weeks.

Discussion

The present study demonstrated that T_{30} is prolonged by vagal blockade but is nearly independent of exercise intensity, exercise hemodynamic variables and sympathetic blockade. These results indicate that the initial heart rate recovery within 30 s is mediated primarily by vagal reactivation and, hence, that T_{30} could be an index of vagally mediated heart rate recovery after exercise. Using this index, we further demonstrated that vagally mediated heart rate recovery is accelerated in well trained athletes but blunted in patients with chronic heart failure.

Rationale of T_{30} as an index of vagal reactivation. Several previous studies (6,7,20) also suggested that time constants of heart rate decay after exercise are mediated by vagal activity. However, Savin et al. (20) observed that the time constant of heart rate recovery at 5 min after intense exercise was also affected by sympathetic blockade. Moreover, Perini et al. (6) reported that the time constant for 5 min after exercise showed a work load dependence, reflecting gradual sympathetic withdrawal after high levels of exercise. In this study, T_{120} also showed a clear work load dependence (Fig. 2). Moreover, the prolongation of T_{120} with atropine was significantly attenuated by concomitant administration of propranolol (Fig. 3). Thus,

Table 3. Effects of Chronic Heart Failure and Endurance Physical Training on Oxygen Uptake and Heart Rate Variables During Exercise

	Age-Matched Control Group for CHF	CHF Group	Age-Matched Control Group for Athletes	Athlete Group
Peak VO_2 (ml/min per kg)	31.0 $\pm$ 5.3	19.5 $\pm$ 2.0*	38.7 $\pm$ 5.0	64.2 $\pm$ 4.5†
VO_2 at AT (ml/min per kg)	23.6 $\pm$ 3.8	15.3 $\pm$ 3.6*	27.3 $\pm$ 5.8	34.1 $\pm$ 7.1‡
Rest HR (beats/min)	78 $\pm$ 12	82 $\pm$ 11	81 $\pm$ 11	66 $\pm$ 5‡
Peak HR (beats/min)	154 $\pm$ 8	142 $\pm$ 23‡	166 $\pm$ 18	178 $\pm$ 10
HR at AT (beats/min)	110 $\pm$ 13	116 $\pm$ 16	123 $\pm$ 14	111 $\pm$ 14
T_{30} at AT (s)	176 $\pm$ 60	426 $\pm$ 250*	126 $\pm$ 43	74 $\pm$ 18†
(range)	(89–290)	(105–1,079)	(75–217)	(59–110)
T_{120} at AT (s)	36 $\pm$ 14	113 $\pm$ 113*	30 $\pm$ 9	20 $\pm$ 7‡
(range)	(18–79)	(20–448)	(13–52)	(13–32)

* $p < 0.01$, † $p < 0.05$ versus control group for chronic heart failure (CHF). ‡ $p < 0.01$, § $p < 0.05$ versus control group for athletes. Data are expressed as mean value $\pm$ SD. Other abbreviations as in Table 2.

because of sympathetic nerve dependence, the time constants of the heart rate decay at ≥ 2 min after exercise could not be used as specific indicators of vagally mediated heart rate recovery. In contrast, T_{30} was nearly independent of exercise intensity and sympathetic blockade (Figs. 2 and 3). Thus, T_{30} could be an index of vagally mediated heart rate recovery after exercise.

Although T_{30} and T_{120} were obtained from the same heart rate decay, they showed different characteristics, depending on the computed models. This result strongly suggests that monoexponential approximation of the overall heart rate decay is not strictly correct. A better fit may be obtained with a biexponential model. However, Perini et al. (6) failed to show that biexponential fitting was superior to monoexponential fitting. Thus, our simple semilogarithmic analysis, independent of the late slow decrease, appears to be a more practical and error-free method for assessing the initial rapid heart rate decay.

Power spectral analysis of heart rate may also be useful in evaluating postexercise vagal reactivation. Arai et al. (8), using this technique, suggested that heart rate recovery at 2 min after exercise is mainly attributed to vagal reactivation. However, this technique requires several consecutive RR intervals over a few minutes for adequate analysis. Thus, because of limited time resolution, power spectral analysis is not appropriate for evaluating rapid and dynamic vagal reactivation after exercise.

Mechanism of vagal reactivation. Although the mechanism leading to prompt vagal reactivation after exercise remains unclear, arterial baroreflex or exercise reflexes originating from mechanosensitive or chemosensitive muscle afferents in exercising legs are not plausible mechanisms because T_{30} depends minimally on the exercise intensity, systolic blood pressure and oxygen uptake at the end of exercise (Fig. 2). Therefore, central mechanisms, such as the release of inhibitory commands from the motor cortex to the parasympathetic center, coupled with cessation of exercise, may play a central role in vagal reactivation.

However, the influence of the somatic exercise reflex cannot be entirely excluded under intense exercise conditions. The T_{30} time constant after maximal exercise was slightly but significantly greater than that after submaximal exercise (Table 2, Fig. 2). Chemosensitive muscle afferents in exercising legs play an important role in the regulation of autonomic nerve activities during exercise (1). After intense exercise, stimulation of muscle afferents with anaerobic metabolites may persist despite cessation of exercise, thereby inhibiting central vagal reactivation. Increased circulating epinephrine may also attenuate vagal reactivation after intense exercise because circulating epinephrine increases the systemic metabolic rate (21). Moreover, extremely activated sympathetic nerves during intense exercise may inhibit neuroeffector transmission of efferent vagal nerves because neuropeptide Y, coreleased with norepinephrine from sympathetic nerve endings, is known to attenuate the inhibitory effect of vagal stimulation (22). Thus, the high sympathetic activity and accumulation of anaerobic metabolites during intense exercise could attenuate postexercise

vagal reactivation. Therefore, submaximal exercise should be used for assessment of the intrinsic potential of vagal reactivation from the initial postexercise heart rate decay. We studied exercise at the anaerobic threshold level in patients with heart failure and in athletes.

Study limitations. This study has four major limitations. First, because of uncomfortable symptoms of vagal blockade, a dose of 0.02 mg/kg of atropine sulfate, which is less than the necessary dose for complete vagal blockade (0.04 mg/kg) (23), was used. However, our finding that T_{30} was markedly prolonged even with partial vagal blockade indicates that the initial heart rate decay is mediated largely by vagal reactivation.

Second, the sequence of the six exercise tests was not randomized in our normal volunteers. Therefore, changes in the time constants of the postexercise heart rate decay relative to exercise conditions may be attributed to a training effect of repetitive exercise performed at 1-week intervals. However, this is unlikely because both T_{30} and T_{120} remained unchanged over 5 weeks during the study.

Third, our patients with chronic heart failure had been treated with digitalis, diuretic agents, angiotensin-converting enzyme inhibitors, nitrates, calcium antagonists and mexiletine. The abnormal heart rate decay in these patients may be attributed partly to drug effects rather than the effect of heart failure. However, vagolytic actions have not been demonstrated in any of these drugs. Conversely, digitalis and angiotensin-converting enzyme inhibitors have vagotonic actions (24-27). Thus, it is unlikely that the abnormal heart rate decay in patients with heart failure is related to the drug effects.

Finally, although vagal activity is known to decrease with advancing age (28), effects of aging on vagally mediated heart rate recovery were not examined in this study. However, the mean age of 56 years in the normal control group for the heart failure group was significantly older than that of 20 years in the control group for the athlete group ($p < 0.05$), and, moreover, T_{30} was significantly greater in the older than in the younger control group (176 ± 60 vs. 126 ± 43 s, $p < 0.001$). These results strongly suggest that vagal reactivation after exercise is blunted with advancing age.

Clinical implications. In the clinical setting, vagal activity is assessed by power spectral analysis of heart rate (9,10) or by arterial or cardiopulmonary baroreflex sensitivity (29,30). The former technique is useful for evaluation of tonic activity at a steady state, whereas the latter reflects reflex vagal activation or deactivation. The time constant of the initial heart rate decay (T_{30}) proposed in this study is likely to reflect dynamic vagal reactivation after exercise, which is mediated by a central command mechanism.

In athletes, both tonic vagal activity and baroreflex sensitivity are increased (17,31). The present study demonstrated that vagally mediated heart rate recovery after exercise is also accelerated in athletes. This may be a physiologic adaptation allowing for rapid heart rate recovery after intense exercise. Although the precise mechanism is unclear, accelerated postexercise vagal reactivation may be associated with an increased

regulatory set point of vagal activity, as reflected by slow rest heart rates (Table 3).

In contrast to athletes, patients with chronic heart failure have decreased tonic vagal activity (11,12) and impaired baroreflex sensitivity (13,14). The present study demonstrated that vagally mediated heart rate recovery is also blunted in these patients. In patients with chronic heart failure, sympathetic activity is often increased during even light exercise (32-34). Moreover, anaerobic metabolites may accumulate even during submaximal exercise in these patients because of abnormalities in skeletal muscle metabolism (35). However, the marked prolongation of T_{30} observed in patients with heart failure is not explained solely by increased sympathetic activity or anaerobic metabolism because in normal volunteers, prolongation of T_{30} was only slight even during maximal exercise (Fig. 2). Thus, blunted vagal reactivation in patients with chronic heart failure may be associated with changes in central command mechanisms. However, it is also possible that end-organ subsensitivities to vagal stimulation attenuate the heart rate response, because Vatner et al. (36) demonstrated a decrease in cardiac muscarinic receptor density in dogs with experimental heart failure. Whatever the mechanism, blunted vagal reactivation may delay postexercise heart rate recovery and potentially exert a harmful effect on the failing heart by increasing myocardial oxygen consumption. The blunted vagal reactivation may also be associated with postexercise lethal ventricular arrhythmias that are often observed in patients with chronic heart failure. Moreover, blunted vagal reactivation may be improved by administration of digitalis or angiotensin-converting enzyme inhibitors and by exercise training, as observed in the athletes. Further studies are necessary to test these hypotheses.

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